5 **(a)** $Q = It = 0.150 \times 20 \times 60 = 180\ \text{C}$

(b) $V = IR = 0.150 \times 12 = 1.8\ \text{V}$

(c) Using the potential divider principle,

$$\frac{12}{12 + R_{eff}} \times 6.0 = 1.8$$

$$12 + R_{eff} = 40$$

$$R_{eff} = 28\ \Omega$$

Since 33 Ω resistor and resistor R are arranged in parallel,

$$\frac{1}{33} + \frac{1}{R} = \frac{1}{28}$$

$$R = 185\ \Omega$$

(d) As the temperature of the thermistor increases, the resistance of the thermistor decreases.
The equivalent resistance across the parallel combination of thermistor and 33 Ω resistor will decrease and the effective resistance of the circuit will decrease.
The ammeter reading will increase.

6 **(a)** $\Phi = BAN$